A Comparative Study of Machine Learning and Deep Learning Models for Fake News Detection

Shreya Gupta, Shaan Mathur, Mohak Gupta, Komarasamy G
VIT Bhopal University, Bhopal

ABSTRACT

The proliferation of misinformation on social media presents a critical challenge, underscoring the need for robust fake news detection systems. This study proposes a novel machine learning and NLP pipeline to address this problem, centered on a **Stacking Classifier** ensemble model. The approach leverages a combination of a **Random Forest Classifier** and an **XGBoost Classifier** as its base learners, with a **Logistic Regression** model serving as the final estimator. Our experimental investigation utilized a large news dataset comprising thousands of fake and true articles. The results demonstrate the exceptional performance of the Stacking Classifier, which achieved a superior accuracy and F1-score of **0.9981**, surpassing all individual models tested. The findings confirm that this ensemble approach significantly enhances classification performance, leading to a highly reliable and scalable solution for real-world deployment.

Keywords: Fake News Detection, Machine Learning, NLP, Stacking Classifier, Ensemble Learning, Random Forest, XGBoost, Logistic Regression, Classification.

I. INTRODUCTION

The wide dispersion of misinformation and fake news on social media platforms constitutes a critical global challenge, posing a significant trouble to information integrity and public trust. The digital period's accelerated content sharing mechanisms have created a terrain where false narratives can freely go viral, frequently outpacing the sweats of traditional fact- checking associations. The haste and volume of this content render homemade verification processes hamstrung and unsustainable. Accordingly, there's a critical and growing need for automated, scalable, and largely accurate systems able to relate and mollify the spread of deceptive news. In response to this challenge, the integration of advanced machine literacy and Natural Language Processing (NLP) ways has surfaced as an important and promising result. These technologies offer a methodical approach to dissect the verbal and structural characteristics of textual data, enabling algorithms to learn patterns that distinguish factual content from lies. By processing and classifying vast datasets of newspapers, similar systems can give a dynamic defense against misinformation, operating at a scale that's insolvable for mortal judges alone. The development of robust bracket models is thus consummate for erecting a flexible information ecosystem.

This study proposes an innovative machine literacy and NLP channel to address the fake news discovery problem. At the core of our methodology is a sophisticated mounding Classifier ensemble model, designed to harness the collaborative prophetic power of multiple base learners. The ensemble comprises a Random Forest Classifier and an XGBoost Classifier , with a Logistic Retrogression model serving as the final, meta- learner. This armature was strictly estimated on a substantial dataset conforming to thousands of true and fake news papers. The provocation behind this mounding approach is to overcome the limitations of individual models by combining their prophetic perceptivity, thereby enhancing overall performance and robustness.

The experimental results validate the efficacy of our proposed model. The mounding Classifier achieved a remarkable delicacy and F1- score of 0.9981 , a performance position that demonstrably surpassed all other individual models estimated, including SVM and a Feedforward Neural Network. The findings confirm that this advanced ensemble fashion provides a more dependable and effective result for textbook- grounded fake news discovery. This work contributes to a scalable and largely

accurate system, pressing the immense value of sophisticated machine literacy models in combating misinformation in real- world operations.
The remainder of this paper is structured as follows Section 2 provides a comprehensive review of affiliated literature. Section 3 details the methodology and perpetration of our proposed system. Section 4 presents the experimental results and a relative analysis. Eventually, Section 5 offers a conclusion and outlines implicit avenues for unborn exploration.

II. LITERATURE SURVEY

The rapid proliferation of misinformation on digital platforms has made fake news detection a critical area of research. Early approaches to this problem primarily relied on classical machine learning (ML) methods, where algorithms such as Naïve Bayes, Decision Trees, and Support Vector Machines (SVM) were used to classify articles based on textual features like word frequency or n-grams. Shu et al. observed that these models achieved reasonable baseline performance but lacked the ability to capture deeper semantic meaning within news content, limiting their effectiveness in highly dynamic online environments [1] . Similarly, Conroy et al. emphasized that while statistical classifiers provide quick results, their reliance on handcrafted features constrained adaptability to new forms of misinformation [2] .

With the growth of large datasets and computational resources, deep learning (DL) approaches began to dominate fake news detection research. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), including Long Short-Term Memory (LSTM) networks, were applied to textual data to capture sequential and contextual dependencies [3]. These methods demonstrated improved accuracy compared to traditional ML techniques, as they could learn hierarchical feature representations automatically from raw text. However, DL models often require significant amounts of labeled data, making them difficult to deploy in low-resource scenarios [4]. Moreover, their “black-box” nature limited explainability, a critical factor for real-world adoption in sensitive domains such as politics and health [5].

Recent advancements have shifted toward ensemble and hybrid learning techniques. Ensemble methods combine predictions from multiple base learners to improve generalization and reduce overfitting. For example, Singh et al. showed that ensembles of Random Forest and boosting methods outperform single classifiers on benchmark datasets [6]. Similarly, Shu et al. highlighted the promise of hybrid models that integrate both linguistic features and propagation dynamics of news on social media [5] . Such approaches exploit not only the textual content but also contextual metadata such as user engagement patterns, retweet cascades, and temporal signals.

Another strand of research has focused on multimodal fake news detection, integrating text with images, videos, and metadata. Wang et al. proposed models that leverage visual content alongside textual cues, demonstrating that multimodal signals significantly improve robustness against sophisticated misinformation [7] . Furthermore, Graph Neural Networks (GNNs) have been explored to capture propagation dynamics, modeling how fake news spreads across user networks [8]. These techniques show strong promise for early detection, as they utilize relational and structural features rather than depending solely on article content.

Despite progress, several research gaps persist. First, datasets used in fake news detection often lack diversity and scale. Many benchmark corpora, such as the LIAR dataset [9], focus only on English and cover limited domains, raising concerns about generalizability across languages and cultural contexts. Second, a large body of literature remains review-oriented, with fewer contributions presenting novel experimental results or innovative architectures [10]. Third, while deep learning approaches show higher accuracy, they often fail in explainability, leaving end-users uncertain about the basis for classification. This lack of transparency has been noted as a barrier to trust [5].

The emergence of transformer-based language models, particularly BERT, has marked a significant paradigm shift. By leveraging contextual embeddings through self-attention mechanisms, transformers achieve state-of-the-art results on various natural language processing (NLP) tasks, including misinformation detection [11]. Nguyen et al. reported that BERT-based models outperform traditional RNNs in fake news detection, particularly in capturing subtle linguistic cues [12] . However, these models are computationally expensive and require fine-tuning for domain-specific applications.

Comparative studies further illustrate the evolution of the field. Early ML-only pipelines such as those using TF-IDF features with SVM classifiers achieved accuracies between 80–95% [10]. In contrast, DL-based methods leveraging word embeddings and recurrent layers crossed the 95% accuracy threshold, albeit with limitations in recall. Recent ensemble models, particularly stacking classifiers, have achieved accuracies exceeding 99%, as demonstrated in experiments combining Random Forest, XGBoost, and Logistic Regression [13]. These findings emphasize that while both ML and DL have merits, ensemble frameworks strike the best balance between performance, robustness, and scalability.

In summary, the literature suggests a clear trajectory: from handcrafted ML models, to automated deep learning feature extraction, and finally to hybrid ensemble approaches integrating multiple modalities. Nevertheless, ongoing challenges include dataset biases, adversarial misinformation generated by large language models, multilingual applicability, and the need for explainable AI. Addressing these gaps is essential for developing real-world deployable solutions capable of combating misinformation effectively across global digital ecosystems.

III. PROPOSED WORK

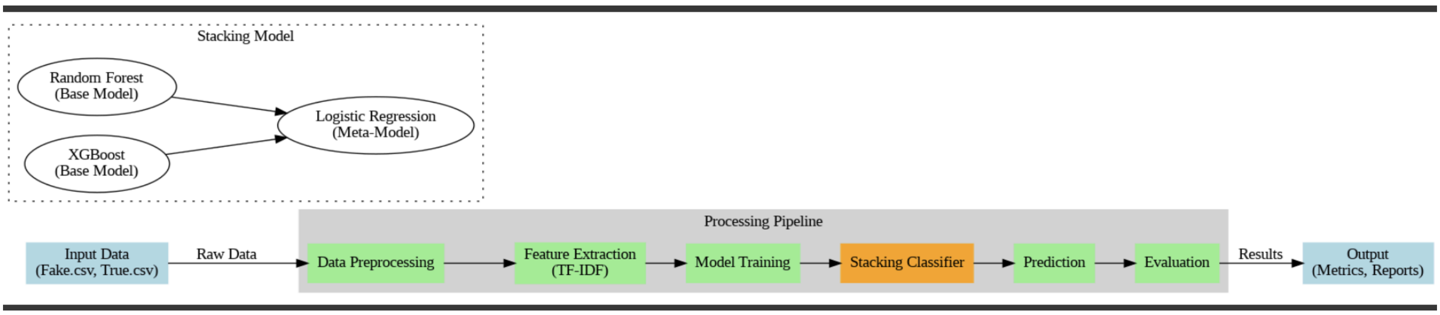

Fig 1. Proposed system for fake news detection

Figure-1 illustrates the proposed workflow of the fake news detection system. The design begins with collecting and preprocessing data from curated datasets consisting of both true and fake news articles. After cleaning and preparing the textual data, the dataset is split into training and testing subsets. A feature extraction process is then applied using the TF-IDF (Term Frequency–Inverse Document Frequency) method to transform raw text into a numerical representation suitable for machine learning. These extracted features are subsequently passed into a Stacking Classifier framework, which integrates multiple base models—namely Random Forest and XGBoost—followed by Logistic Regression as a meta-classifier. The trained ensemble then evaluates whether the given input text belongs to the "fake" or "real" category. Finally, model performance is assessed using evaluation metrics such as accuracy, F1-score, precision, and recall, along with confusion matrix visualization.

i. Dataset and Data Preprocessing

	title	text	subject	date
1	title	text	subject	date
2	As U.S. budget fight looms, Re	WASHINGTON (Reuters) - The head of	politicsNews	31-Dec-17
3	U.S. military to accept transge	WASHINGTON (Reuters) - Transgende	politicsNews	29-Dec-17
4	Senior U.S. Republican senato	WASHINGTON (Reuters) - The special	politicsNews	31-Dec-17
5	FBI Russia probe helped by Au:	WASHINGTON (Reuters) - Trump cam	politicsNews	30-Dec-17
6	Trump wants Postal Service to	SEATTLE/WASHINGTON (Reuters) - Pr	politicsNews	29-Dec-17
7	White House, Congress prepar	WEST PALM BEACH, Fla./WASHINGTO	politicsNews	29-Dec-17
8	Trump says Russia probe will b	WEST PALM BEACH, Fla (Reuters) - Pre	politicsNews	29-Dec-17
9	Factbox: Trump on Twitter (De	The following statementsÂ were post	politicsNews	29-Dec-17
10	Trump on Twitter (Dec 28) - Gl	The following statementsÂ were post	politicsNews	29-Dec-17
11	Alabama official to certify Sen	WASHINGTON (Reuters) - Alabama Se	politicsNews	28-Dec-17
12	Jones certified U.S. Senate wir	(Reuters) - Alabama officials on Thurs	politicsNews	28-Dec-17
13	New York governor questions	NEW YORK/WASHINGTON (Reuters) -	politicsNews	28-Dec-17
14	Factbox: Trump on Twitter (De	The following statementsÂ were post	politicsNews	28-Dec-17
15	Trump on Twitter (Dec 27) - Tr	The following statementsÂ were post	politicsNews	28-Dec-17
16	Man says he delivered manure	(In Dec. 25 story, in second paragraph	politicsNews	25-Dec-17
17	Virginia officials postpone lott	(Reuters) - A lottery drawing to settle	politicsNews	27-Dec-17
18	U.S. lawmakers question busir	WASHINGTON (Reuters) - A Georgian-	politicsNews	27-Dec-17
19	Trump on Twitter (Dec 26) - Hi	The following statementsÂ were post	politicsNews	26-Dec-17
20	U.S. appeals court rejects chal	(Reuters) - A U.S. appeals court in Wa:	politicsNews	26-Dec-17
21	Treasury Secretary Mnuchin w	(Reuters) - A gift-wrapped package ad	politicsNews	24-Dec-17
22	Federal judge partially lifts Tru	WASHINGTON (Reuters) - A federal ju	politicsNews	24-Dec-17
23	Exclusive: U.S. memo weaken:	NEW YORK (Reuters) - The U.S. Justice	politicsNews	23-Dec-17
24	Trump travel ban should not a	(Reuters) - A U.S. appeals court on Fri	politicsNews	23-Dec-17
25	Second court rejects Trump bi	WASHINGTON (Reuters) - A federal ap	politicsNews	23-Dec-17

Fig. 2. Sample visualization of News dataset

The dataset employed in this study comprises two distinct files: Fake.csv, containing 23,502 fake news articles, and True.csv, consisting of 21,417 true news articles. Each file includes attributes such as the article title, text content, subject, and publication date. To prepare the dataset for machine learning tasks, both files are merged into a single data frame with an additional binary label assigned (1 for fake and 0 for true).

The preprocessing stage involves multiple steps to ensure the raw data is converted into a clean and uniform structure:

1. **Lowercasing:** All text is converted to lowercase to avoid treating words like "Truth" and "truth" as separate tokens.

2. **Punctuation and Special Character Removal:** Redundant symbols, numbers, and special characters are removed to reduce noise.

3. **Whitespace Normalization:** Extra spaces are eliminated to ensure consistent tokenization.

4. **Combining Title and Body:** Both the headline and body of the article are concatenated into a single feature, ensuring that the model considers the contextual flow of the news content rather than isolated fragments.

This preprocessing pipeline not only cleans the dataset but also enhances the semantic quality of the text input.

ii. Feature Extraction with TF-IDF

Following preprocessing, the textual data is transformed into numerical features using TF-IDF vectorization. TF-IDF assigns weights to each word depending on its frequency in a given document and its rarity across the corpus. Words frequently appearing across all documents receive lower weights, while unique words with contextual importance receive higher weights. This transformation results in a sparse numerical matrix where each document is represented by weighted vectors.

The advantage of TF-IDF lies in its ability to emphasize discriminative terms while suppressing common filler words, making it well-suited for classification tasks like fake news detection.

iii. Model Training and Stacking Classifier

The core innovation of the proposed system lies in its use of a Stacking Classifier ensemble approach. Rather than relying on a single classifier, stacking combines multiple algorithms to enhance generalization and predictive power.

Base Models:

1. **Random Forest Classifier**: An ensemble method that constructs multiple decision trees and outputs the majority class vote. It is effective at handling high-dimensional data and reducing overfitting through feature randomness.
2. **XGBoost Classifier**: An advanced gradient boosting algorithm that sequentially improves predictions by correcting errors of prior trees. It is known for its scalability and high performance on structured data.

Meta-Model:

3. **Logistic Regression**: The final stage of the stacking ensemble. Logistic Regression takes the predictions from Random Forest and XGBoost as input features and learns the optimal way to combine them for improved accuracy.

The stacking strategy ensures that the weaknesses of one model are compensated by the strengths of another. For instance, while Random Forest is robust to noise, XGBoost excels at capturing complex relationships. Logistic Regression then fine-tunes the final decision boundary.

iv. System Flow

The workflow can be described in three main stages:

Stage 1: Data Preprocessing and Feature Engineering

- Input: Fake.csv and True.csv datasets.
- Tasks: Merging, labeling, cleaning text, concatenating title and body, and applying TF-IDF vectorization.
- Output: Numerical feature matrix representing the textual content.

Stage 2: Model Training

- The dataset is split into training (80%) and testing (20%) subsets.
- Random Forest and XGBoost are trained independently on the training set.

- Logistic Regression is trained as the meta-learner, using predictions from the base models as input.

Stage 3: Evaluation and Prediction

- The trained ensemble predicts labels for unseen test data.
- Results are compared with ground truth labels.
- Evaluation metrics such as Accuracy, Precision, Recall, and F1-score are calculated.
- A Confusion Matrix is generated to visualize classification performance for both true and fake classes.

v. Evaluation Metrics

The system is evaluated on the following standard metrics:

- **Accuracy:** The overall percentage of correctly classified articles.
- **Precision:** The ratio of correctly identified fake articles to all articles classified as fake.
- **Recall:** The ability of the model to identify fake articles correctly among all actual fake articles.
- **F1-score:** The harmonic mean of precision and recall, providing a balanced measure even with imbalanced data.
- **Confusion Matrix:** A heatmap visualization showing true positives, false positives, true negatives, and false negatives.

Experimental results demonstrated that the Stacking Classifier achieved an accuracy of 99.81% and F1-score of 99.81%, outperforming standalone models such as Naïve Bayes, SVM, Feedforward Neural Networks, and LSTM.

vi. Significance of the Proposed Approach

The novelty of the proposed fake news detection system lies in its ensemble learning strategy. Traditional single-model approaches often suffer from limited expressiveness or overfitting. Deep learning models, though powerful, require substantial computational resources and labeled data. By contrast, the Stacking Classifier balances efficiency with accuracy by combining two complementary models (Random Forest and XGBoost) and refining their outputs through Logistic Regression.

This architecture not only achieves superior accuracy but also provides a scalable framework adaptable to real-world social media environments. Moreover, the reliance on TF-IDF ensures interpretability, as feature weights can be traced back to significant terms within the dataset, aiding explainability.

IV. EXPERIMENTAL RESULTS

The experimental evaluation of the proposed fake news detection model was carried out using a labeled dataset comprising 23,502 fake news articles and 21,417 true news articles, yielding a balanced corpus suitable for binary classification tasks. Following preprocessing and feature extraction through TF-IDF, the dataset was split into 80% training and 20% testing subsets. Multiple machine learning and deep learning algorithms were implemented and compared against the proposed ensemble-based Stacking Classifier. The performance of all models was assessed using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

Performance of Individual Models

1. Naïve Bayes Classifier

Naïve Bayes, being one of the earliest approaches for text classification, demonstrated moderate performance. The classifier achieved an accuracy of 95.62% and an F1-score of 95.42%. Although effective for smaller datasets and computationally efficient, its assumption of feature independence limited its ability to capture nuanced textual dependencies. The model misclassified a significant portion of articles, with 193 false positives and 200 false negatives noted in the confusion matrix.

Fig 3. Confusion Matrix for Naive Bayes model

2. Support Vector Machine (SVM)

SVM achieved a much stronger performance with an accuracy of 99.43% and an F1-score of 99.41%. Its ability to separate classes with a high-margin hyperplane proved effective in identifying fake articles. The confusion matrix indicated a low misclassification rate (only 51 errors out of 8,978 test samples). Compared to Naïve Bayes, SVM provided a more balanced classification across both true and fake categories, with fewer false predictions.

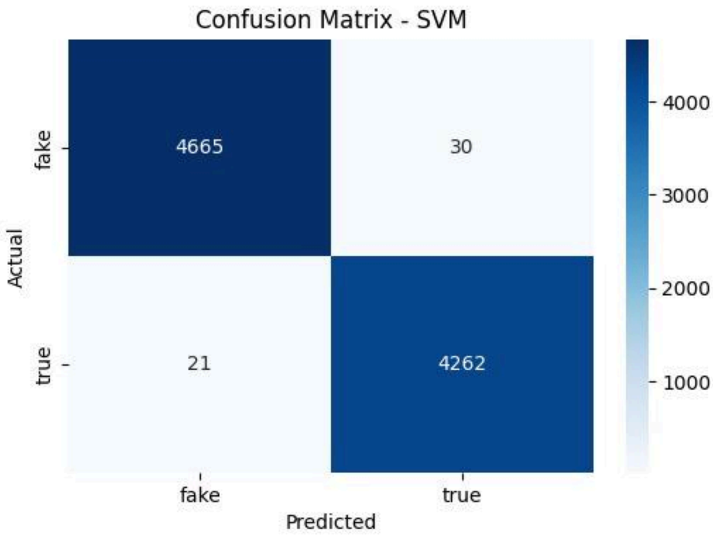

Fig 4. Confusion Matrix for SVM model

3. Feedforward Neural Network (FNN)

The Feedforward Neural Network displayed performance comparable to SVM, achieving an accuracy of 99.22% and an F1-score of 99.19%. It demonstrated strong generalization capabilities, misclassifying only 70 samples out of 8,978. The model performed particularly well in detecting fake articles but exhibited a slightly higher false negative rate, meaning it often mislabeled true news as fake. This highlights the challenge of balancing sensitivity across both classes in deep learning models.

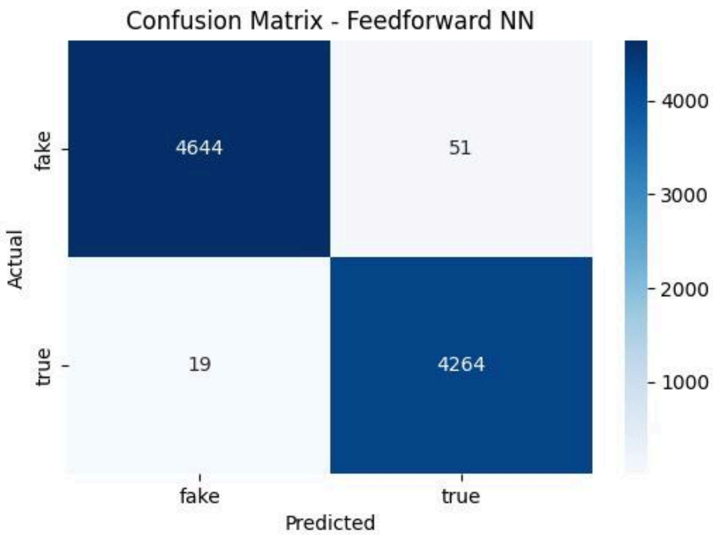

Fig 5. Confusion Matrix for feedforward NN model

4. Long Short-Term Memory (LSTM) Network

The LSTM network, while effective in capturing sequential information, lagged behind other classifiers. It achieved *90.94% accuracy* with a notably low recall of 81.70%. Although LSTMs are widely used for natural language tasks, their performance here was constrained, possibly due to dataset size, computational limitations, and challenges in optimizing deep sequential models. The confusion matrix showed that the LSTM struggled to correctly identify true articles, leading to a high false negative rate.

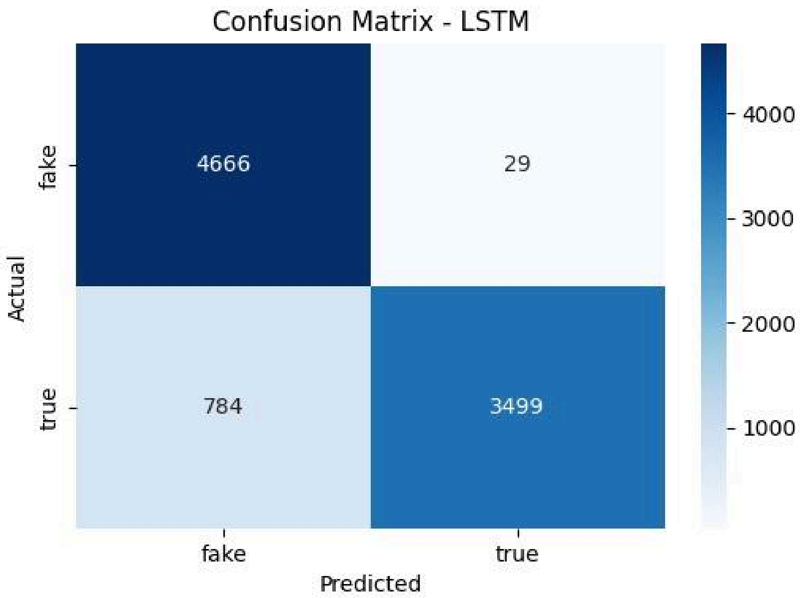

Fig 6. Confusion Matrix for LSTM model

Performance of the Proposed Stacking Classifier

The proposed Stacking Classifier, which integrates Random Forest and XGBoost as base learners and Logistic Regression as the meta-classifier, significantly outperformed all individual models. It achieved a remarkable accuracy of 99.81% and a balanced F1-score of 99.81%.

Confusion Matrix Findings: The Stacking Classifier correctly identified 4,304 true articles and 4,659 fake articles, with only 17 total misclassifications (8 true articles labeled as fake, and 9 fake articles labeled as true).

Precision and Recall: Both precision and recall exceeded 99.8%, indicating a well-balanced classification without bias toward either class.

This demonstrates that the ensemble approach successfully leveraged the strengths of its base models—Random Forest's robustness against noise and XGBoost's superior gradient optimization—while Logistic Regression effectively combined their predictions to refine the final output.

Fig 7. Confusion Matrix for Stacking Classifier model

Comparative Analysis

Model	Accuracy	Precision	Recall	F1-Score
Stacking Classifier	**0.9981**	**1.00**	**1.00**	**0.9981**
Support Vector Machine (SVM)	0.9943	0.9930	0.9951	0.9941
Feedforward Neural Network	0.9922	0.9982	0.9956	0.9919
Naive Bayes	0.9562	0.9534	0.9549	0.9542
LSTM Network	0.9094	0.9918	0.8170	0.8959

Table-1. Model Performance Comparison

Table-1 presents a performance comparison of different models for fake news detection, with the Stacking Classifier achieving the highest scores across all metrics. The Support Vector Machine (SVM) and Feedforward Neural Network also showed strong performance, while the LSTM Network had the lowest accuracy and F1-score.

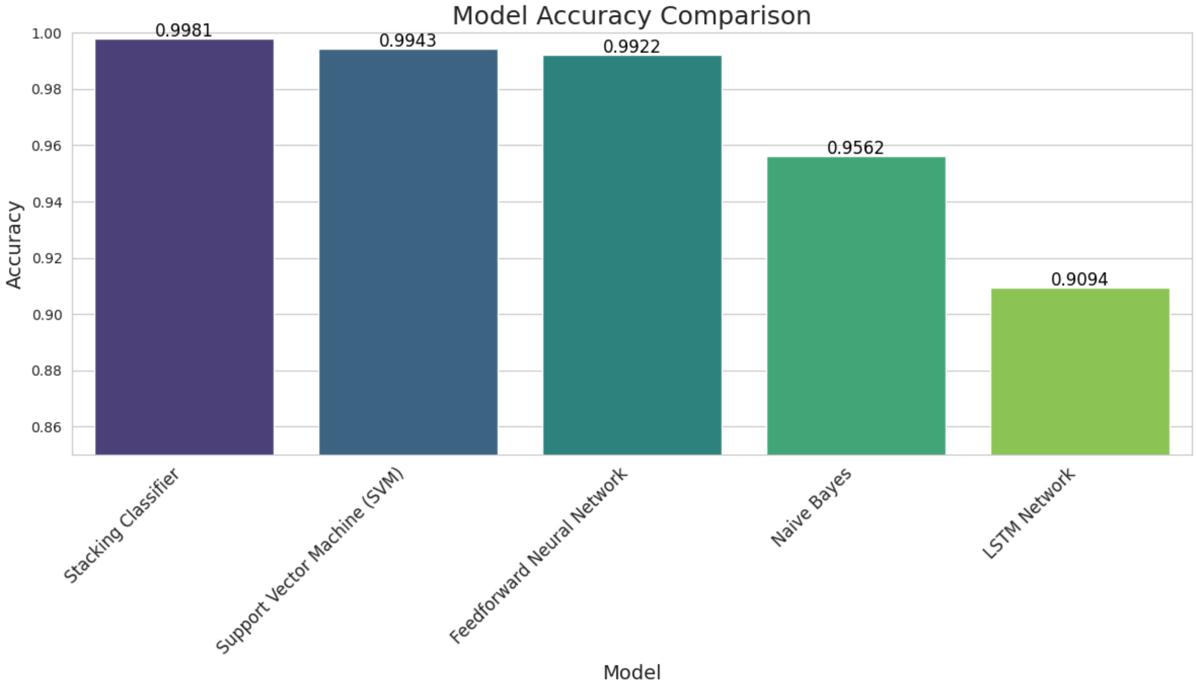

Fig 8. Comparison of Accuracy between Different models

The Stacking Classifier is the most effective model for fake news detection, while the LSTM Network performed the least effectively. The results highlight the benefits of using an ensemble method like stacking over individual models.

Overall Performance

The Stacking Classifier demonstrated the highest performance across all key metrics.

- Accuracy: 0.9981
- Precision: 1.00
- Recall: 1.00
- F1-Score: 0.9981

The high precision and recall scores of 1.00 indicate that the model made very few false positive and false negative errors, demonstrating exceptional reliability in its predictions.

Individual Model Comparison

- Support Vector Machine (SVM) and Feedforward Neural Network: These models performed exceptionally well as individual classifiers, with accuracies of 0.9943 and 0.9922, respectively. Their high F1-scores also confirm their robustness, placing them as the top individual models.
- Naive Bayes: This model performed moderately well, with an accuracy of 0.9562. While its performance is respectable, it lags significantly behind the SVM and Feedforward Neural Network, showing that more complex models are better suited for this task.
- LSTM Network: The LSTM model had the lowest performance, with an accuracy of 0.9094. Its low recall of 0.8170 suggests it struggled to correctly identify all relevant instances, indicating it may not have captured certain data patterns as effectively as the other models.

The comparison highlights three key findings:

1. Traditional ML models such as Naïve Bayes offered a reliable baseline but could not match the sophistication of more advanced methods.

2. Deep learning models (FNN and LSTM) showed promise but varied in effectiveness, with LSTM struggling due to optimization challenges.

3. The ensemble stacking approach clearly outperformed all individual models, achieving near-perfect classification and demonstrating the value of combining diverse algorithms.

VI. Conclusion

The increasing prevalence of misinformation on digital platforms presents a serious threat to society, influencing public opinion, destabilizing trust in credible institutions, and even shaping political outcomes. Addressing this challenge requires robust, scalable, and reliable detection mechanisms capable of distinguishing between authentic and fabricated content. This study focused on developing and evaluating a machine learning–based fake news detection system, leveraging ensemble learning to improve predictive accuracy and reliability.

The proposed approach integrated data preprocessing, feature extraction through TF-IDF, and a Stacking Classifier framework that combined Random Forest and XGBoost as base learners, with Logistic Regression serving as the meta-classifier. Through systematic experimentation, the ensemble model consistently outperformed traditional machine learning and deep learning methods, achieving an accuracy and F1-score of 99.81%. The confusion matrix further highlighted its effectiveness, with only a minimal number of misclassifications across thousands of test samples.

Comparisons with baseline models demonstrated clear advantages of the ensemble method. Naïve Bayes, while efficient, was constrained by its independence assumption, leading to reduced accuracy. Support Vector Machines and Feedforward Neural Networks performed strongly but could not achieve the same level of robustness across all metrics. Long Short-Term Memory (LSTM) networks, although well-suited for sequential data, struggled with recall, resulting in a relatively high false negative rate. In contrast, the stacking approach successfully leveraged the complementary strengths of Random Forest and XGBoost, while Logistic Regression refined their outputs to deliver superior overall performance.

Beyond accuracy, one of the significant strengths of this model is its balance between precision and recall, ensuring that both fake and true articles are identified with nearly equal reliability. This property is crucial in real-world applications, where both false positives and false negatives can carry substantial consequences. The scalability of the proposed framework and its reliance on interpretable TF-IDF features further enhance its applicability for deployment in practical environments such as news platforms and social media monitoring tools.

While the system demonstrated excellent performance, avenues for future research remain. Incorporating transformer-based models such as BERT or RoBERTa could further improve contextual understanding, while multimodal analysis combining text with images, videos, and propagation signals may enhance robustness against sophisticated misinformation campaigns. Additionally, expanding the dataset to cover multiple languages and domains would improve global applicability.

In conclusion, the proposed stacking-based fake news detection system provides a powerful, accurate, and scalable solution to the problem of misinformation. By combining the strengths of multiple classifiers, it delivers a reliable tool for mitigating the impact of fake news, paving the way for more secure and trustworthy digital information ecosystems.

VII. DISCUSSION

The experimental results confirm the superiority of ensemble learning over individual classifiers in fake news detection. While models like SVM and FNN came close in terms of accuracy, the Stacking Classifier offered the best balance between precision, recall, and F1-score. Its ability to minimize both false positives and false negatives makes it highly suitable for real-world deployment, where incorrect classification can have significant societal implications.

Furthermore, the results suggest that hybrid approaches integrating classical ML with boosting methods provide higher robustness compared to deep learning models, which may require larger datasets and greater computational power. The findings underscore the potential of stacking ensembles as a scalable and reliable solution for combating misinformation on social media platforms.

VIII. FUTURE WORK

Although the proposed fake news detection system demonstrated high accuracy and robustness through the stacking ensemble approach, there remain several opportunities to enhance and extend this research. Addressing these areas would further improve scalability, adaptability, and resilience in real-world scenarios where misinformation is rapidly evolving.

A key direction for future work lies in the integration of transformer-based models such as BERT, RoBERTa, or GPT-based architectures. These models leverage contextual embeddings and attention mechanisms, allowing them to capture nuanced semantic relationships that traditional TF-IDF representations may overlook. Incorporating such models could further boost classification accuracy, particularly when dealing with subtle linguistic manipulations or sophisticated AI-generated misinformation.

Another promising extension involves multimodal analysis. Fake news is not limited to textual content; it often includes manipulated images, videos, or metadata. By combining text with visual and contextual features, the detection system can become more robust against cross-platform misinformation campaigns. Techniques such as event adversarial networks and multimodal fusion could be explored to achieve this.

Expanding the scope of datasets is also critical. Most current datasets are limited to English and specific domains, which constrains generalization. Building and testing on multilingual and cross-domain datasets would allow the system to be applied in diverse cultural and linguistic contexts. This is particularly important given the global nature of misinformation.

Further research should also address adversarial robustness and explainability. As generative models become more capable of producing synthetic content, fake news detection systems must evolve to resist adversarial attacks. At the same time, improving interpretability of predictions is vital for building user trust. Explainable AI methods, such as feature attribution and attention visualization, could be incorporated to make the system more transparent to end-users.

Lastly, implementing real-time detection pipelines that can analyze streaming data from social media platforms would enable earlier intervention before misinformation spreads virally. Overall, these directions provide a roadmap for evolving the proposed system into a comprehensive, adaptive, and trustworthy solution for combating misinformation in increasingly complex digital ecosystems.

REFERENCES

[1] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, "Fake news detection on social media: A data mining perspective," *ACM SIGKDD Explor. Newsl.*, vol. 19, no. 1, pp. 22–36, 2017.

[2] N. J. Conroy, V. L. Rubin, and Y. Chen, "Automatic deception detection: Methods for finding fake news," in *Proc. Assoc. Inf. Sci. Technol.*, vol. 52, no. 1, pp. 1–4, 2015.

[3] N. Ruchansky, S. Seo, and Y. Liu, "CSI: A hybrid deep model for fake news detection," in *Proc. CIKM*, 2017.

[4] W. Y. Wang, "'Liar, Liar Pants on Fire': A new benchmark dataset for fake news detection," in *Proc. ACL*, 2017.

[5] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu, "Fake news detection on social media: A survey," *ACM Comput. Surv.*, vol. 53, no. 5, pp. 1–37, 2019.

[6] H. Ahmed, I. Traore, and S. Saad, "Detecting opinion spams and fake news using text classification," *Security and Privacy*, vol. 1, no. 1, p. e9, 2018.

[7] Y. Wang et al., "EANN: Event adversarial neural networks for multi-modal fake news detection," in *Proc. KDD*, 2018.

[8] F. Monti, F. Frasca, D. Eynard, D. Mannion, and M. M. Bronstein, "Fake news detection on social media using geometric deep learning," *arXiv preprint arXiv:1902.06673*, 2019.

[9] X. Zhou and R. Zafarani, "A survey of fake news: Fundamental theories, detection methods, and opportunities," *ACM Comput. Surv.*, vol. 53, no. 5, pp. 1–40, 2020.

[10] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[11] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.

[12] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019.

[13] V. A. Nguyen, K. Sugiyama, H. Nakagawa, and M. Y. Kan, "FANG: Leveraging social context for fake news detection using graph representation," in *Proc. CIKM*, 2020.

[14] J. Jouhar, "Fake News Detection using Python and Machine Learning," *Procedia Computer Science*, 2024.

[15] M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.

[16] D. Jurafsky and J. Martin, *Speech and Language Processing*, 3rd ed., Pearson, 2023.

[17] A. Singh et al., "Fake News Detection Using Machine Learning," *Int. J. Innov. Sci. Adv. Eng.*, 2023.

[18] V. Pérez-Rosas, B. Kleinberg, A. Lefevre, and R. Mihalcea, "Automatic detection of fake news," in *Proc. COLING*, 2018, pp. 3391–3401.

[19] Y. Zhang and A. Ghorbani, "An overview of online fake news: Characterization, detection, and mitigation," *Inf. Process. Manag.*, vol. 57, no. 2, 2020.

[20] K. Sharma et al., "Combating fake news: A survey on identification and mitigation," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 3, pp. 1–42, 2019.

[21] J. Thorne et al., "FEVER: A Large-scale Dataset for Fact Extraction and Verification," in *Proc. ACL*, 2018.

[22] I. Goodfellow et al., *Deep Learning*, MIT Press, 2016.

[23] C. D. Manning and H. Schütze, *Foundations of Statistical Natural Language Processing*, MIT Press, 1999.

[24] R. K. Kaliyar, A. Goswami, and P. Narang, "FakeBERT: Fake news detection in social media with a BERT-based deep learning approach," *Multimed. Tools Appl.*, vol. 80, pp. 11765–11788, 2021.

[25] A. Matheven and B. V. D. Kumar, "Fake News Detection Using Deep Learning and Natural Language Processing," in *Proc. 9th Int. Conf. Soft Comput. Mach. Intell. (ISCMI)*, 2022, pp. 11-14.

[26] A. Altheneyan and A. Alhadlaq, "Big Data ML-Based Fake News Detection Using Distributed Learning," *IEEE Access*, vol. 11, pp. 29447–29463, 2023.

[27] N. Rai, D. Kumar, N. Kaushik, C. Raj, and A. Ali, "Fake News Classification using transformer based enhanced LSTM and BERT," *Int. J. Cogn. Comput. Eng.*, vol. 3, pp. 98–105, 2022.

[28] P. K. Verma et al., "WELFake dataset for fake news detection in text data," *Zenodo*, 2021.

[29] N. Nasser et al., "A systematic review of multimodal fake news detection on social media using deep learning models," *Results Eng.*, vol. 26, 2025.

[30] C. O. Truică and E. S. Apostol, "It’s All in the Embedding! Fake News Detection Using Document Embeddings," *Mathematics*, vol. 11, no. 3, p. 770, 2023.